MILITARY SPECIFICATION

ANALYZER, ENGINE, ELECTRONIC, CONSOLE-TYPE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one type of engine analyzer used to test automotive internal combustion, spark-ignition engines of commercial and military vehicles.

2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-566	-	Box, Folding, Paperboard
PPP-B-621	-	Box, Wood, Nailed and Lock Corner
PPP-B-636	-	Box, Shipping, Fiberboard
PPP-C-1797	-	Cushioning Material, Cellulosic
PPP-T-60	-	Tape, Packaging, Waterproof
PPP-T-97	-	Tape, Pressure-Sensitive Adhesive, Filament Reinforced

MILITARY

MIL-P-116	-	Preservation, Methods of

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Munitions and Chemical Command, ATTN: DRSAR-LEE-S(R), Rock Island, IL 61299 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4910

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MIL-B-121 - Barrier Material, Greaseproofed, Waterproofed, Flexible
MIL-T-152 - Treatment, Moisture-and Fungus-Resistant, of Communications, Electronic, and Associated Electrical Equipment

STANDARDS

FEDERAL

FED-STD-H28 - Screw Thread Standards For Federal Services

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-275 - Printed Wiring for Electronic Equipment
MIL-STD-454 - Standard General Requirements for Electronic Equipment
MIL-STD-889 - Dissimilar Metals
MIL-STD-1186 - Cushioning, Anchoring, Bracing, Blocking, and Waterproofing; With Appropriate Test Methods

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

DRAWINGS

ARMY

AL

7540877 - Adapter, Primary Circuit
7540884 - Adapter, Spark Plug
7540895 - Adapter, Ignition Circuit
7541200 - Adapter, Testing Regulator
7541209 - Adapter, Testing Generator
7541281 - Adapter set, Wheeled Vehicle

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DOD adopted shall be the issue listed in the current DODISS and the supplement thereto, if applicable.

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NATIONAL ELECTRICAL MANUFACTURERS ASSOCIATION (NEMA)

Electrical Standards

(Application for copies should be addressed to the National Electrical Manufacturers Association, 155 East 44th Street, New York, New York 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D3951-82 - Standard Practice for Commercial Packaging

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

* 2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 First article. When specified (see 6.2), the contractor shall furnish one complete analyzer for first article inspection. The first article may be either a preproduction model or an initial production item which conforms to the requirements of this specification. In either case, the approved first article and the production items shall be identical and in accordance with the terms of the contract. Approval of the first article shall not relieve the contractor of the responsibility to furnish equipment in accordance with the requirements of this specification.

* 3.2 Design. The analyzer shall be new, mobile, and meet the performance requirements as specified herein. The analyzer shall be designed specifically for automotive-engine diagnosis, tuneup, and service. The analyzer shall test two, three, four, six and eight cylinder, two and four cycle engines having 12- and 24-volt negative ground electrical systems. The analyzer shall test both conventional distributor breaker point and solid state, breakerless, high energy ignition systems, including capacitive discharge types. The analyzer shall operate from 115 volts, 60 hertz, single-phase power. The analyzer shall be a console-type, mounted on a roll-around stand provided with inclosed storage cabinet with doors.

* 3.2.1 Military vehicles. The engine analyzer shall meet all of the requirements of this specification when used on military vehicles with 24-volt ignition systems including the M151A1, 1/4-ton, 4 x 4 utility truck or the M151A2 truck. The electrical engine adapter set 7541281 may be used to interface the analyzer with the 24-volt M151 (military jeep) ignition system. The electrical engine adapter set consists of the following parts:

Adapter, Primary Circuit - Army Drawing B7540877
Adapter, Adapter Spark Plug - B7540884
Adapter, Adapter Ignition Unit - B7540895

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Adapter, Testing, Regulator - B7541200
Adapter, Testing, Generator - C7541209

The electrical engine adapter set is not to be supplied as part of the engine analyzer and the Army Drawings are herein provided for information only.

* 3.2.2 Maintainability. Instructions for disassembly, repair, reassembly, and adjustments shall be clear, concise, and definitive in application. All parts subject to wear, breakage, or distortion shall be accessible for adjustment, repair, or replacement.

3.2.3 Safety provisions. Each analyzer shall incorporate safety devices to prevent conditions hazardous to personnel or detrimental to equipment. The analyzer shall be grounded to prevent electrical shock. When required (see 6.2), special safety devices shall be as specified by the procuring activity.

3.2.4 Interchangeability. All replaceable parts shall be designed and manufactured to such standards as shall permit replacement and adjustment without modification of the parts or the analyzer.

* 3.3 Materials. All materials used in the fabrication of the analyzer shall be new and of the quality necessary to meet the requirements of this specification. Where dissimilar metals are in contact, especially brass, copper, or steel in contact with aluminum or magnesium, they shall be protected against corrosion in accordance with MIL-STD-889.

* 3.4 Construction. The analyzer shall be furnished assembled and complete with all accessories so that it can be used for all operations for which it was intended.

3.4.1 Electrical connections. Connections of conductors and terminal parts shall be of the screw, pressure, or solder-type, except as specified herein. When soldered connections are used, the conductors and terminal parts shall be mechanically secured before soldering. Soldering shall be in accordance with MIL-STD-454, Requirement 5. Only rosin base fluxes shall be used in soldering operations. Connections to screw-type terminals shall be mechanically secured with means to prevent loss of tightness. Printed circuitry shall be in accordance with MIL-STD-275.

3.4.1.1 Electrical equipment. Unless otherwise specified (see 6.2), all electrical components including motors, starters, relays, switches, wiring, and their locations shall conform to the applicable NEMA electrical standards.

3.4.2 Castings and forgings. All castings shall be free from blow-holes, porosity, hard spots, shrinkage defects, cracks, or other defects. Forgings shall be free from scale, inclusions, cold shuts, mismatching, sharp edges, or other defects. Castings or forgings shall be free from any properties adversely affecting their strength, durability, or suitability.

3.4.3 Welding, brazing, or soldering. Welding, brazing or soldering shall be employed only where such operations are a requirement of the original design and shall not be employed in the repair of any defective part.

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3.4.4 Threads. Threads shall conform to FED-STD-H28 and the applicable Detailed Standard section referenced therein.

3.5 Performance characteristics.

* 3.5.1 Accuracy. With the analyzer at an ambient air temperature of +75 degrees F ±10 degrees F, the error of the meters (in their respective and associated circuits) and gages (except for combustion efficiency tester, ohmmeter, and timing advance meter) shall not exceed 3.0 percent of their full-scale value.

3.5.1.1 Calibration. All meters, gages and measuring devices shall be adjustable to allow periodic recalibration. The meters and gages shall be initially calibrated against certified standards before delivery to the Government.

3.5.2 Test display stability. The oscilloscope shall be designed to maintain a stable pattern at any engine speed within the operating range of 500 to 3000 revolutions per minute (rpm) without manual control or adjustment. The oscilloscope pattern movement shall not be more than ±5.0 percent of the usable screen diameter in the vertical or horizontal direction after 15 minutes of warm up. For the purpose of this requirement, the synchronizing device's signal and the pattern generating device (magneto or distributor) shall have no "jitter" incidental to engine gear backlash. Patterns on the oscilloscope shall be clear and essentially free from deflections due to stray voltages and fields.

3.5.3 Temperature and humidity conditions.

3.5.3.1 Operating temperature. The analyzer and all components shall operate within the accuracy requirements of 3.5.1 and without damage in ambient temperatures ranging from +30 degrees F to +115 degrees F.

3.5.3.2 Humidity conditions. The analyzer shall withstand the following conditions without damage and shall operate within the accuracy requirements of 3.5.1 after:

(a) twenty hours exposure to an air temperature of +75 degrees F to +77 degrees F, with a relative humidity of 93 to 97 percent, and with air movement and radiation negligible.
(b) four hours exposure to an air temperature of +110 degrees F to +115 degrees F, with a relative humidity of 20 to 25 percent, and with air movement and radiation negligible.

* 3.5.3.3 Storage conditions. When specified (see 6.2), the analyzer shall withstand the following conditions and shall operate with the accuracy requirements of 3.5.1 after:

(a) 72 hours of exposure to -65°F ± 10°F.
(b) 24 hours of exposure to +95°F ± 10°F.
(c) 4 hours of exposure to +165°F ± 10°F.

3.6 Details of components. All components shall be of the best commercial quality.

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3.6.1 Meters, gages, and digital displays. All round gages shall have a diameter of at least 2.5 inches and a scale arc length of at least 240 degrees. All rectangular meters (except for timing advance meter, when mounted in the timing light) shall be at least 6.1 inches in width and shall have an indicator scale length of at least 4.8 inches. Meters with digital displays may be substituted for analog meters, provided they have the same or better accuracy as required in 3.5.1. Digital displays shall be at least 3-1/2 digits and have nonblinking readout characters at least 0.4 inches in height that can be easily read from a distance of at least 10 feet. Digital meters shall be accurate to within 0.1 percent of reading $\pm$ one digit exclusive of associated circuitry. All multifunction meters, gages, and digital displays shall have a function selector, and all multiscale meters, gages, and digital displays shall have a scale selector. All panel-mounted meters and gages shall be illuminated and easily readable.

3.6.1.1 Tachometer. The tachometer shall be capable of measuring engine rpm over a range of at least 350 to 7,500 rpm. Analog meters shall indicate on two or more scales. The lower rpm scale shall indicate over a range of at least 350 to 1,250 rpm, but shall not exceed a range of 0 to 7,500 rpm. The lower rpm scale shall be incremented in intervals no larger than 25 rpm. The scale(s) indicating the upper range of rpm shall be incremented in intervals no larger than 200 rpm.

3.6.1.2 Dwell indicator. The dwell indicator device shall be capable of measuring dwell angle on at least 2-, 3-, 4-, 6-, and 8-cylinder engines. Analog meters shall indicate dwell angle over a range of 0 to at least 90 degrees on one or more scales. The scales shall be incremented in intervals no larger than 1 degree.

* 3.6.1.3 Vacuum pressure gage. This unit shall measure vacuum and pressure in the fuel system. The vacuum pressure gage shall have scales capable of measuring vacuum from zero to 30 inches of mercury and pressure from zero to five pounds per square inch (psi). All accessories for connecting the tester to the engine shall be furnished.

* 3.6.1.4 Fuel pump volume flow and pressure gages. The volume flow shall be measured in pints per minute in one ounce minimum graduation readings. Minimum of one pint graduated measuring container shall be provided. The fuel pump pressure shall be measured in psi on a gage scale of 0 to at least 10 psi. The gage error shall not exceed 3.0 percent of full scale value at any point on the scale.

3.6.1.5 Voltmeter. The voltmeter shall be capable of testing primary voltages within a range of at least 0 to 40 volts direct current. The indicator may be either a panel meter or a digital readout. Secondary voltages may be indicated on the oscilloscope. The analyzer circuit shall automatically select the correct polarity.

3.6.1.6 Ohmmeter. The ohmmeter shall be capable of measuring resistance over a range of at least zero to 100,000 ohms. The indicator may be either a panel meter or a digital display. The panel meter shall indicate accurately within 2 percent of the linear arc length at the "infinity" marking with open external circuits.

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* 3.6.2 Oscilloscope. The oscilloscope shall have at least a 12-inch (diagonal) screen and shall have an actual useable display pattern width of at least 7.5 inches. It shall have two kilovolt (kv) scales. One shall be marked with a range from 0 to 20 kv or higher and the other shall be marked with a range from 0 to 40 kv or higher. It shall have horizontal scales for measuring dwell variation of 2-, 3-, 4-, 6-, and 8-cylinder engines. The oscilloscope shall display the following patterns of both primary and secondary ignition circuits:

Single cylinder
Parade (20-30 kv and 40-50 kv)
Superimposed
Stacked (raster)
Alternator circuit

After warmup, the oscilloscope shall maintain a stable pattern at any engine speed between 500 and 3,000 rpm without manual control or adjustment. Vertical and horizontal pattern movement (drift) shall not exceed ± 5 percent of the useable display grid height or width.

3.6.2.1 Test display. The presentation, as produced by ignition events, shall be obtained by either primary or secondary ignition circuit pickup, or both, as required by the specific circuitry of the oscilloscope. The displays shall be clear, well defined, and stable. The displays shall permit determination of the total available voltage from the coil, the voltage required to overcome rotor and spark plug gaps, and the variable resistance characteristics in the engine secondary circuitry. The displays shall present sufficient information to permit location and identification of the following ignition circuit events, with the engine operating under various load conditions throughout their respective speed ranges.

(a) Open or shorted portions of ignition circuits resulting from faulty insulation, flashover, defective coils, capacitors, and/or ignition harness.
(b) Leaking, fouled, or shorted spark plugs.
(c) Arcing or bouncing breaker points.
(d) Defective cam, camshaft, rotor, bearings, or cam follower (magneto and distributor) affecting point opening, point closing, and dwell in distributor systems.

3.6.2.2 Vertical gain control (height). A control shall be provided to adjust the amplitude of the vertical deflections produced by the ignition signals. The controlled gain shall be high enough to amplify the vertical displacement sufficient to extend at least 2/3 of the cathode ray tube (crt) face.

3.6.2.3 Pattern expand control (expand). A control shall be provided to control the gain of the horizontal circuits so that sweep length may be adjusted to any point within the range delineated by the following:

(a) Presentation of a single trace across the crt, which portrays engine ignition events occurring during a complete engine cycle of 720 degrees (four-cycle engine).

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(b) Presentation of a portion of the complete 720-degree trace horizontally expanded across the crt. The control shall allow the expanded portion to cover a crankshaft rotation of at least 130 degrees, each degree to be represented by a distance of 1/32 inch along the base line. The point of trace initiation shall not move more than 1/2 inch when sweep length is changed from a single sweep presentation to the expanded portion thereof, or vice versa, within an engine speed range of 500 to 3,000 rpm.

3.6.2.4 Horizontal centering control (parade). A control shall be provided for horizontal centering of the complete 720-degree trace. The control shall provide for parading the entire expanded trace across the face of the crt to allow viewing ignition patterns related to the individual cylinders.

3.6.2.5 Vertical centering control. A control shall be provided to adjust the vertical position of the crt sweep to any position between the top and bottom of the crt screen.

3.6.2.6 Focus control. A manual or automatic control shall be provided to focus the oscilloscope beam.

3.6.2.7 Beam intensity control. A manual or automatic control shall be provided to adjust the intensity of the oscilloscope beam.

3.6.2.8 Raster. The oscilloscope shall stack primary and secondary ignition patterns for isolation and identification of each cylinder. A control/selector switch shall be provided to select the raster function.

3.6.2.9 Superimposed patterns. The oscilloscope shall superimpose primary and also secondary ignition system patterns. A control/selector switch shall be provided to select the superimpose function.

* 3.6.3 Emission analyzer. When specified (see 6.2), the engine analyzer shall be furnished with a means of measuring hydrocarbons (HC) and carbon monoxide (CO) present in the engine exhaust gases.

* 3.6.4 Timing light. The timing light shall be designed to measure initial timing and ignition advance of internal combustion engines. The timing light shall provide sufficient illumination to read and measure initial timing and ignition advance when the marked engine dynamic balancer and the stationary timing pointer are in direct sunlight with a cloudless atmosphere. The timing light shall be suited for operation in any position without missing, double-flashing, or failure. The timing light shall be powered by and attached to the engine analyzer circuitry. The timing light shall incorporate an integral timing advance meter for measuring centrifugal (mechanical) or vacuum advance and the combination of vacuum plus centrifugal advance. The advance meter shall have a range of at least 60 degrees, incremented in no larger than 1 degree intervals. The meter shall indicate the ignition advance accurate to within 5 percent of the full scale value.

3.6.5 Remote control switch. The remote control starter ignition switch shall provide ignition and starter motor control from under the hood positions. Provisions shall be made for either operating the starter motor with the ignition off or on, controllable from the remote switch.

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* 3.6.6 Engine kill switch. The analyzer shall be provided with a switch or button for shorting out the ignition system of all cylinders for test purposes and for stopping the engine quickly.

3.6.7 Top dead center indicator. This indicator shall indicate the crankshaft angle with relation to top dead center for ignition and valve timing purposes and will locate piston top dead center.

* 3.6.8 Cylinder compression tester. A cylinder compression tester with threaded adapters for 10-mm, 14-mm short and long reach with flat and tapered seats, 18-mm with flat and tapered seats, and 7/8 inch flat seat spark plug ports shall be furnished with each engine analyzer unit. The compression tester shall incorporate a dial indicating bourdon tube-type gage with transparent oil and gasoline resistant plastic crystal with a threaded-retaining bezel. The gage housing shall provide automatic pressure relief in case of bourdon tube failure due to overpressure. The compression tester shall contain a check valve that retains the pressure admitted to the gage until released by the operator. A manually operated relief valve, accessible from the exterior of the gage or gage body, which releases the retained pressure and returns the gage pointer to zero shall be provided. The relief valve shall be operable without disassembly of the gage, without removal of the gage from the spark plug hole, and without tools.

* 3.6.9 Engine power balance tester. A panel-mounted, engine-running, power-balance tester, with individual cylinder test switches or buttons, shall be located on the console panel to permit electrical shorting of the ignition of individual cylinders under test. Weak cylinders shall be identified by the absence or a marked change of engine rpm.

* 3.6.10 Test leads and accessories. Leads for connection of the vehicle to the analyzer shall be routed through the test harness boom. These leads shall be of the clamp-on or clamp-around type so that there are no breaks in vehicle connections. Other necessary leads, adapters, etc that are normally provided with the manufacturer's commercial analyzer shall be furnished. Indirect, inductive, or Hall effect pickup connections to the engine shall be fully insulated to prevent direct electrical connection.

3.6.11 High tension pliers. Each analyzer shall be provided with high tension pliers for handling "hot" spark plug wires without danger of electrical shock to the operator.

3.6.12 Stand. The stand shall be capable of bearing the combined weight of all test equipment and accessories without any evidence of strain, distortion, or deformity. The stand shall consist of one unit divided into two parts. The lower part shall be an enclosed cabinet with storage facilities for accessories or components not mounted on the stand. The upper part shall provide facilities for supporting or mounting the components of the analyzer in a position so they can be easily used. The analyzer stand shall be provided with four standard swivel casters at least four inches in diameter. All four casters shall be provided with foot-operated locks. Both analyzer and analyzer stand shall have hand holders capable of supporting the full weight of the analyzer/analyzer stand.

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3.7 Fungus control. Unless otherwise specified (see 6.2), electrical components and circuit elements shall be treated for fungus resistance in accordance with type II requirements of MIL-T-152, except that the statement of treatment is not required.

* 3.8 Painting. Painting and finishing of the analyzer shall be in accordance with the best commercial practice within the engine analyzer console manufacturing industry.

* 3.9 Marking. The analyzer shall be identified with the following information marked on a plastic (mylar) or metal film plate:

NOMENCLATURE:
NSN:
CONTRACT NUMBER:
MFR NAME AND ADDRESS:
VOLTS:
HERTZ:
PHASE:

3.10 Dielectric strength. The analyzer shall show no evidence of smoking, arcing, flashover, charred insulation, or other damage when subjected to the application of 500 volts root mean square (rms) for a period of one minute between mutually insulated portions or between insulated portions and ground. The secondary lead circuit shall be subjected to 30,000 volts (zero to peak) for a period of one minute. When an external pickup is used, only this pickup shall be subjected to the application of 30,000 volts (zero to peak) between the pickup body and ground.

* 3.11 Workmanship. Standards of workmanship shall assure that the engine analyzer shall have the efficient operating characteristics found in standard commercial units and as specified in Section 3 herein.

* 4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the suppliers may use their own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) First article inspection (see 4.2.1).
(b) Quality conformance inspection (see 4.2.2).

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4.2.1 First article inspection. When specified (see 6.2), first article inspection shall be performed on the preproduction model(s) or initial production item(s) noted in 3.1. First article inspection shall consist of the examination in 4.3 and all tests under 4.4. Failure of the first article to pass the examination or any of the tests shall be cause for rejection.

4.2.2 Quality conformance inspection.

4.2.2.1 Inspections and tests. Quality conformance inspection shall be applied to production units offered for acceptance under the contract. Quality conformance inspection shall consist of (a) through (c) as follows, and failure of any item to pass an examination or test shall be cause for rejection of the item.

(a) Product examination (see 4.3).
(b) Accuracy test (see 4.4.1) and oscilloscope stability test (see 4.4.2).
(c) Packaging inspection (see 4.5).

4.2.2.2 Sampling. Sampling for quality conformance inspection shall be performed in accordance with MIL-STD-105 using the following:

Inspection or test	Level	AQL
Product examination	II	1.0
**Accuracy test	II	1.0
Packaging inspection	S-4	4.0

*Maximum lot size of the accuracy test shall not exceed 150 units.

4.3 Product examination. The trestles shall be examined to determine compliance with the requirements of Section 3, using the following checklist as a guide.

CHARACTERISTICS	REQUIREMENT PARAGRAPH
Design	3.2
Maintainability	3.2.2
Safety provisions	3.2.3
Interchangeability1/	3.2.4
Materials1/	3.3
Construction	3.4
Electrical connections1/	3.4.1
Electrical equipment1/	3.4.1.1
Castings and forgings	3.4.2
Welding, brazing, or soldering	3.4.3
Threads1/	3.4.4
Meters, gages, and digital displays	3.6.1
Tachometer	3.6.1.1
Dwell indicator	3.6.1.2
Vacuum pressure gage	3.6.1.3
Fuel pump volume flow and pressure gages	3.6.1.4
Voltmeter	3.6.1.5

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Ohmmeter	3.6.1.6
Oscilloscope	3.6.2
Test display	3.6.2.1
Vertical gain control	3.6.2.2
Pattern expand control	3.6.2.3
Horizontal centering control	3.6.2.4
Vertical centering control	3.6.2.5
Focus control	3.6.2.6
Beam intensity control	3.6.2.7
Raster	3.6.2.8
Superimposed patterns	3.6.2.9
Emission analyzer (when specified)	3.6.3
Timing light1/	3.6.4
Remote control switch	3.6.5
Engine kill switch	3.6.6
Top dead center indicator	3.6.7
Cylinder compression tester	3.6.8
Engine power balance tester	3.6.9
Test leads and accessories	3.6.10
High tension pliers	3.6.11
Stand	3.6.12
Fungus control (when specified)1/	3.7
Painting	3.8
Marking	3.9
Workmanship	3.11

1/A certificate of conformance may be provided to satisfy the examination requirement where more than a visual or dimensional verification is necessary.

4.4 Tests.

4.4.1 Accuracy test. The following analog and digital meters, gages and timing light shall be tested in their respective circuits to determine compliance with paragraph 3.5.1. Tests will be conducted at +75°F ±10°F with reference meters and gages having at least four times specified accuracy. Failure of any analog and digital meter, gage, or timing light to provide reading to specified accuracy or any timing light failing to operate as specified shall be cause for rejection. The test method shall be at the option of the supplier. All analog and digital meters and gages shall be calibrated according to paragraph 3.5.1.1 prior to this test.

4.4.1.1 Tachometer accuracy test. Operate the tachometer through a range of 350 to 7500 rpm. Analog meter error shall not exceed 3.0 percent of the linear length at any point on the scale (see 3.6.1.1).

4.4.1.2 Dwell indicator accuracy test. Operate the dwell indicator device through a range of from 0° to 90°. Analog meter error shall not exceed 3.0 percent of the linear length at any point on the scale (see 3.6.1.2).

4.4.1.3 Vacuum pressure gage accuracy test. Operate the vacuum-pressure gage from zero to 30 inches of mercury and pressure from zero to five pounds per square inch (psi). Error of gage readings shall not exceed 3.0 percent of the linear arc length at any point on the scale (see 3.6.1.3).

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* 4.4.1.4 Fuel pump volume flow and pressure gage tests.

(a) Operate the fuel pump volume flow system to measure fuel flow in pints per minute. Graduations of one ounce minimum for accurate readings.
(b) Operate the fuel pump pressure gage through a range from 0 to 10 psi. The error in the gage readings shall not exceed 3.0 percent of full scale value at any point (see 3.6.1.4).

* 4.4.1.5 Emission Analyzer Test. Operate the emission analyzer and read the engine emission of hydrocarbons (HC) and Carbon Monoxide, error shall not exceed 3.0 percent of the full scale value on the gages at any point.

4.4.1.6 Voltmeter test. Operate the voltmeter through a range of 0 to 40 volts direct current. The error of the analog meter or gage shall not exceed 3.0 percent of the linear length/linear arc length at any point on the scale. Digital meters shall be accurate to within 0.1 percent of reading plus or minus one count (see 3.6.1.5).

4.4.1.7 Ohmmeter test. Operate the ohmmeter through a range of zero to 100,000 ohms. If a panel meter is used, it shall indicate accurately within 2 percent of the linear arc length at the "infinity" marking with open external circuits. If a digital display is used, the digital meter shall be accurate to within 0.1 percent of reading plus or minus one count (see 3.6.1.6).

4.4.1.8 Timing light test. The timing light shall be operated to verify that it functions (light measurement readings are not required as part of this test). The timing light shall operate without skipping or double-flashing (see 3.6.4).

4.4.2 Oscilloscope stability test. This test shall be performed at an ambient temperature of +75° ±10°F after a 15 minute warm-up period of the oscilloscope. Any 2, 3, 4, 6 or 8 cycle engine (or as specified in para 3.2) or equivalent mockup representative of the ignition system type shall be used. The oscilloscope shall maintain a stable pattern at any engine speed within the operating range of 500 to 3000 revolutions per minute (rpm) without manual control or adjustment. The oscilloscope pattern movement shall not be more than ±5.0 percent of the usable screen diameter in the vertical or horizontal direction. Patterns on the oscilloscope shall be clear and free from deflections due to stray voltages and fields. No "jitter" shall be present in the signal of the pattern generating device due to engine gear backlash. Failure to meet the requirements of paragraph 3.5.2 shall be cause for rejection.

4.4.3 Operating temperature test. The analyzer shall be cooled to a temperature of +32 degrees ±two degrees F and, while stabilized at that temperature, tested in accordance with 4.4.1 and 4.4.2. The analyzer shall then be heated to a temperature of +113 degrees F ±two degrees F and, while stabilized at that temperature, tested in accordance with 4.4.1 and 4.4.2. Failure of the engine analyzer to pass the test of 4.4.1 and 4.4.2 at either temperature, or evidence of damage to any part of the analyzer due to the temperature extremes shall be cause for rejection (see 3.5.3.1).

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* 4.4.3.1 Storage conditions test. When specified (see 6.2), the analyzer shall be placed in an ambient air temperature of -65°F ±10°F for 72 hours. At the end of this time the analyzer shall be placed in an ambient air temperature of +95°F ±10°F for a period of 24 hours. The temperature shall then be increased over a 5 hour period to +165°F and held for four hours at +165°F ±10°F. The temperature shall then be steadily decreased and allowed to stabilize at an ambient air temperature of +75°F ±10°F. The analyzer shall then be inspected for evidence of damage and tested for accuracy in accordance with 3.5.1.

4.4.4 Humidity test. The analyzer shall be exposed for twenty hours to an air temperature of +75 to +77 degrees F, with a relative humidity of 93 to 97 percent, and with air movement and radiation negligible. While stabilized at that condition, the analyzer shall be tested in accordance with 4.4.1 and 4.4.2. Following the above, the analyzer shall be exposed for four hours to an air temperature of +110 to +115 degrees F, with a relative humidity of 20 to 25 percent, and with air movement and radiation negligible. While stabilized at that condition, the analyzer shall again be tested in accordance with 4.4.1 and 4.4.2. Failure of the analyzer to pass the test of 4.4.1 and 4.4.2 or evidence of corrosion or other damage shall be cause for rejection (see 3.3 and 3.5.3.2).

4.4.5 Performance test. The analyzer shall be turned on and allowed to warm up for fifteen minutes. Following the warmup period, the oscilloscope controls shall be operated to verify compliance to the requirements of 3.6.2.2 through 3.6.2.9, and the oscilloscope display checked for stability. The analyzer shall then be used to perform engine analysis of 2- and 4- cycle engines or equivalent mockups representative of the ignition system types specified in 3.2 and 3.2.1. For each ignition system tested, the oscilloscope shall correctly portray a representative pattern of each of the applicable ignition events specified in 3.6.2 and 3.6.2.1, and all applicable meter readings shall be taken. Portrayals obtained shall be checked for accuracy by comparison with sample patterns. During the test, the proper functioning of the timing light, remote control switch, engine kill switch, top dead center indicator, cylinder condition analyzer, engine power balance tester, and the emission analyzer (when specified, see 6.2) shall be demonstrated. Failure of any analyzer control, meter, or accessory to function as specified; failure of the analyzer display to correctly portray each applicable ignition event for each ignition system tested; or failure of the oscilloscope display to remain stable and essentially free of deflections from stray voltages and fields shall be cause for rejection (see 3.2, 3.2.1, 3.5.2, and 3.6.1.1 thru 3.6.10).

4.4.6 Dielectric strength test. The analyzer shall be subjected to application of 500 volts (RMS) for a period of one minute between mutually insulated portions and between mutually insulated portions and ground. The secondary lead circuit or external pickup shall be subjected to 30,000 volts (zero to peak) for one minute. The analyzer shall then be inspected for damage and tested in accordance with 4.4.1 and 4.4.2. Failure of the analyzer to withstand the dielectric strength test without evidence of smoking, arcing, flash-over, charred insulation, or other damage; or failure of the analyzer to pass the test of 4.4.1 and 4.4.2 shall be cause for rejection (see 3.5.1 and 3.10).

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4.5 Packaging inspection. The engine analyzer shall be inspected before and after packaging to determine compliance with the preservation, packaging, packing and marking requirements of Section 5.

5. PACKAGING

* 5.1 Preservation. Preservation shall be Level A or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The analyzer and accessories shall be cleaned and dried in accordance with process C-1 and dried in accordance with procedure D-1 or D-4 of MIL-P-116.

5.1.1.2 Preservative application. All unpainted ferrous metal surfaces of the storage cabinet shall be preserved with type P-7 preservative. Caster bearings shall be preserved with type P-11 preservative. Preservatives shall conform to the requirements of MIL-P-116.

* 5.1.1.3 Unit packs. The preserved surfaces shall be wrapped with barrier material conforming to MIL-B-121, grade A. Secure barrier in place with tape conforming to PPP-T-60, type III Class I. The control panel shall be covered with cushioning material conforming to PPP-C-1797, Type I, and secured with the tape specified above. The casters, timing light, and fuel pump tester shall be wrapped with cushioning material and individually packaged in paperboard boxes conforming to PPP-B-566, variety 2. All remaining accessories shall be wrapped with cushioning material. The cushioning material shall be as specified above.

* 5.1.1.3.1 Consolidated packaging. Accessories shall be consolidated into one or more fiberboard container(s) conforming to PPP-B-636, class weather-resistant. Secure the packaged accessories to a shelf within the storage cabinet, utilizing filament tape conforming to PPP-T-97, type optional. The unit method of preservation for the control panel shall conform to method IIa or method IIb of MIL-P-116. The storage cabinet shall be preserved in accordance with method I of MIL-P-116. It shall be acceptable if the supplier elects to preserve the complete assembly method IIa.

* 5.1.2 Commercial. Commercial preservation shall be as specified in ASTM D3951-82.

* 5.2 Packing. Packing shall be Level A or Commercial as specified (see 6.2).

5.2.1 Level A. Each analyzer shall be packed in a wood box conforming to PPP-B-621, class 2, style 2. Blocking, bracing, and cushioning shall be in accordance with MIL-STD-1186. Closure shall be in accordance with the appendix of the box specification.

* 5.2.2 Commercial. Commercial packing shall be as specified in ASTM D3951-82.

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* 5.3 Marking. Marking shall be Level A or Industrial as specified (see 6.2).

* 5.3.1 Level A marking. Level A marking shall be as specified in MIL-STD-129.

* 5.3.2 Commercial marking. Industrial marking shall be as specified in ASTM D3951-82.

6. NOTES

6.1 Intended use. The analyzer covered by this specification is intended for use in automotive repair shops for testing and diagnosing reciprocating, internal combustion, spark-ignition engines.

6.2 Acquisition requirements. Acquisition documents should specify the following:

(a) Title, number, and date of this specification.
(b) First article when required (see 3.1).
(c) Special safety devices when required (see 3.2.3).
(d) Electrical requirements if different (see 3.4.1.1).
(e) Emission analyzer when required (see 3.6.3).
(f) Fungus control when required (see 3.7).
(g) First article inspection if different (see 4.2.1).
(h) Storage conditions when specified (see 3.5.3.3, 4.4.3.1).
(i) Level of preservation, packing, and marking required (see 5.1, 5.2, and 5.3).

6.3 Safety and health requirements. In order that equipment integrated into the user's operational environment will comply with OSHA limitations and control of noise levels, radiation, electromagnetic emission, noxious vapors, heat, etc, as applicable, specific requirements concerning such points of operation, and other health and safety requirements, should be specified by the user.

6.4 Supersession data. This specification includes the requirements of Rock Island Arsenal Purchase Description T & E PD-57, dated 1 August 1968.

6.5 Changes from previous issue. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and suppliers are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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